

# Nels Martin

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## EDUCATION

### University of Washington

Sep 2022 – June 2026

*Bachelor of Science in Electrical and Computer Engineering*

Seattle, WA

- **Coursework:** Data Structures and Algorithms, Intro to Database Systems, Digital Systems, Machine Learning for Signal Processing, Data Driven Machine Learning, Computer-Aided Reasoning for Software Engineering.

## EXPERIENCE

### Helm AI

March 2026 – Present

*Software Development Engineering Intern*

Bellevue, WA

- Building a digital twin simulation platform to support strategic decision making.

### UW Lean Research Group

March 2025 – Present

*Research Assistant*

Seattle, WA

- Designed and implemented an end-to-end reinforcement learning system that interfaces with the Lean 4 theorem prover via the Language Server Protocol (LSP).
- Led development of a policy-gradient (REINFORCE) agent in PyTorch to dynamically suggest proof actions.
- *Technologies:* Python, PyTorch, JSON, Lean 4 Theorem Prover

### Korry Electronics

June 2024 – Sep 2024

*Electrical Engineering Intern*

Everett, WA

- Resolved 100+ engineering problem reports by building test circuits and debugging designs.
- Communicated across departments (engineering, manufacturing, purchasing, etc.) to reduce resolution time.
- *Technologies:* Simetrix, PADs, Xpedition

### nLight

July 2023 – Sep 2023

*Electrical and Computer Engineering Intern*

Camas, WA

- Programmed and calibrated prototype LiDAR cameras in Python, LabVIEW, and Linux.
- Delivered four functional prototypes used in investor presentations, directly supporting product strategy.
- *Technologies:* Python, LabVIEW, Linux.

## PROJECTS

### AutogramGenerator.com

Sep 2025 – Nov 2025

- Built the first website that finds autogramic pangrams: sentences that describe their own letter counts.
- Developed front end with React and Typescript using the Next.js framework, and hosted website from Vercel.
- Wrote back end in Rosette, and deployed the program in a Docker container hosted on Google Cloud Run.
- *Technologies:* React, Typescript, Next.js, Vercel, Racket, Rosette, Docker, Google Cloud Run

### Correctness Checking and Memory Planning for ML Workloads

Sep 2025 – Dec 2025

- Collaborated to design a domain-specific language (DSL) to express training pipelines and model architectures of ML workloads as part of the final group project for CSE 507: Computer-Aided Reasoning for Software Engineering.
- *Technologies:* Python, PyTorch, Racket, Rosette, Git, GitHub

## LEADERSHIP & AWARDS

### Engineering Peer Educator

Sep 2025 – Dec 2025

- Mentored first-year engineering undeclared students through their transition to UW Engineering by facilitating the GEN ST 199 component of Engineering FIGs.

### Emerging Leaders in Engineering Scholarship

Sep 2025 – June 2026

- Selected for a competitive, donor-funded leadership program developing skills in public speaking, mentorship, and navigating ambiguity; received up to \$1,800 in scholarships.

## SKILLS

**Languages :** Python, PyTorch, JavaScript (React.js), Lean, Racket, Rosette, SQL, Java

**Tools :** Git, GitHub, Next.js, Vercel, Docker, Google Cloud, Linux